

Chapter 6 – Non-Seasonal ARIMA Model

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August 2021

1 The ARIMA Model

1. We now look at non-seasonal ARIMA models.
2. The I in the ARIMA stands for “integrated” so the whole thing ARIMA model stands for *Autogressive Integrated Moving Average* model.
3. It is probably best to illustrate with an example, so we will use the austres dataset in **R** .
4. Example 1. Fitting an ARIMA model to the austres data set. This is the number of Australian residents measured quarterly in thousands from 2nd quarter 1971 to 2nd quarter 1993 inclusive.

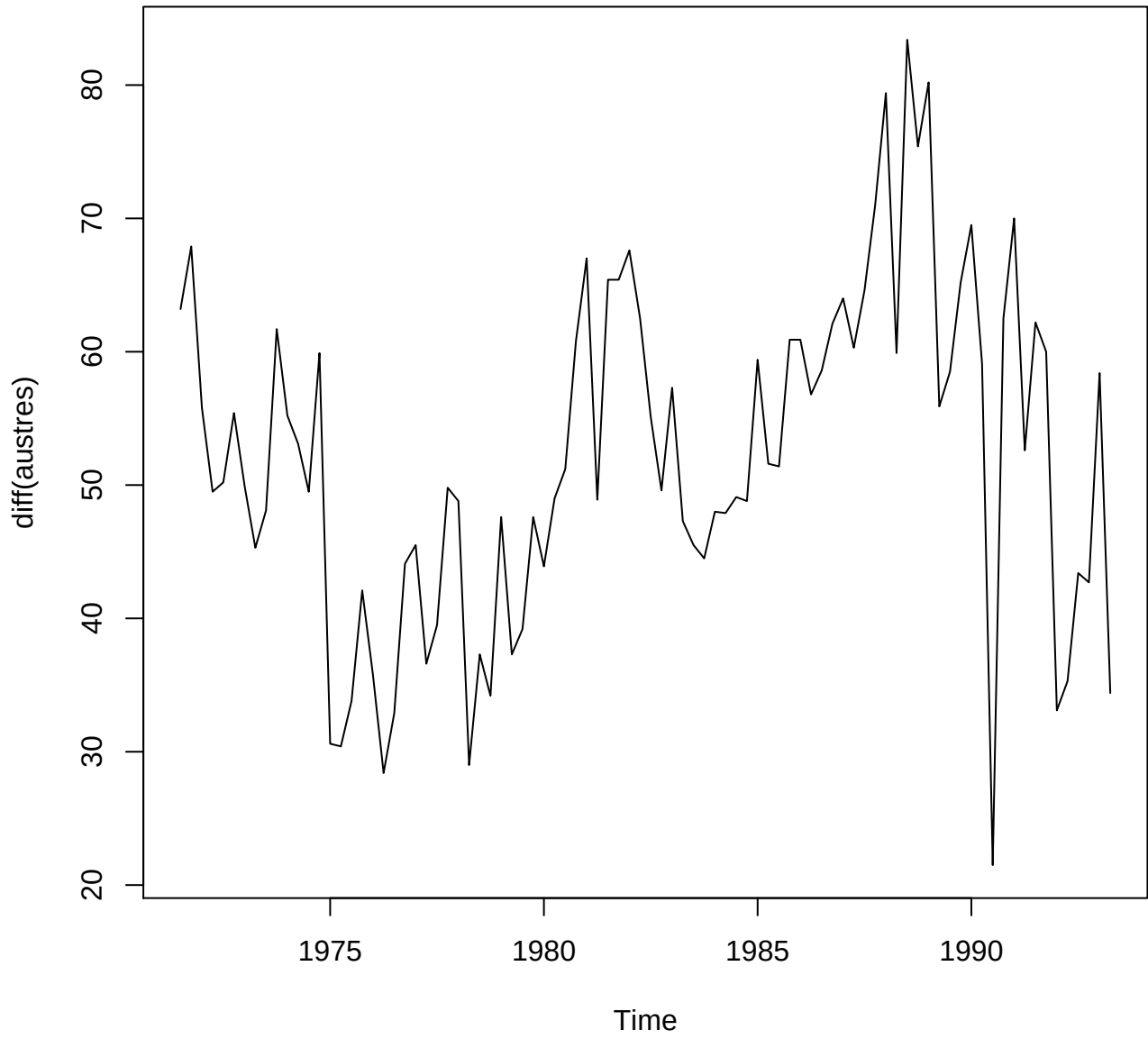
We first look at the plots of austres, and then the first order difference and then the 2nd order difference. Let x_t be the t th observation in the t th quarter where 2nd quarter of 1971 is $t = 1$.

```
> plot(austres)
> plot(diff(austres))
```

Clearly the original plot shows an increasing trend that could possibly be linear. So we look at the 1st order difference plot, that is, the plot of ∇x_t . The plot looks stationary. See Figure 1 for the plot of ∇x_t . For convenience, we write $y_t = \nabla x_t$.

We now look at the correlogram and partial correlogram of $y_t = \nabla x_t$.

Figure 1: Plot of 1st Order Difference of Austres



```
> acf(diff(austres),lag.max=40)
> pacf(diff(austres),lag.max=40)
```

You can check these plots of the correlogram and partial correlogram yourself.

The partial correlogram of ∇x_t shows significant lags up to lag 3, so we can try fitting an ARIMA(3,1,0) to x_t . Note that when we fit an ARIMA(3,1,0) model to x_t , it is the same as fitting an AR(3) model to $y_t = \nabla x_t$.

```
> arima(austres, order=c(3,1,0))
```

Call:

```
arima(x = austres, order = c(3, 1, 0))
```

Coefficients:

	ar1	ar2	ar3
	0.5531	0.0957	0.3411
s.e.	0.1019	0.1207	0.1030

```
sigma^2 estimated as 102: log likelihood = -330.2, aic = 668.4
```

We check the correlogram and partial correlogram of the residuals of our initial fit (see Figure 2).

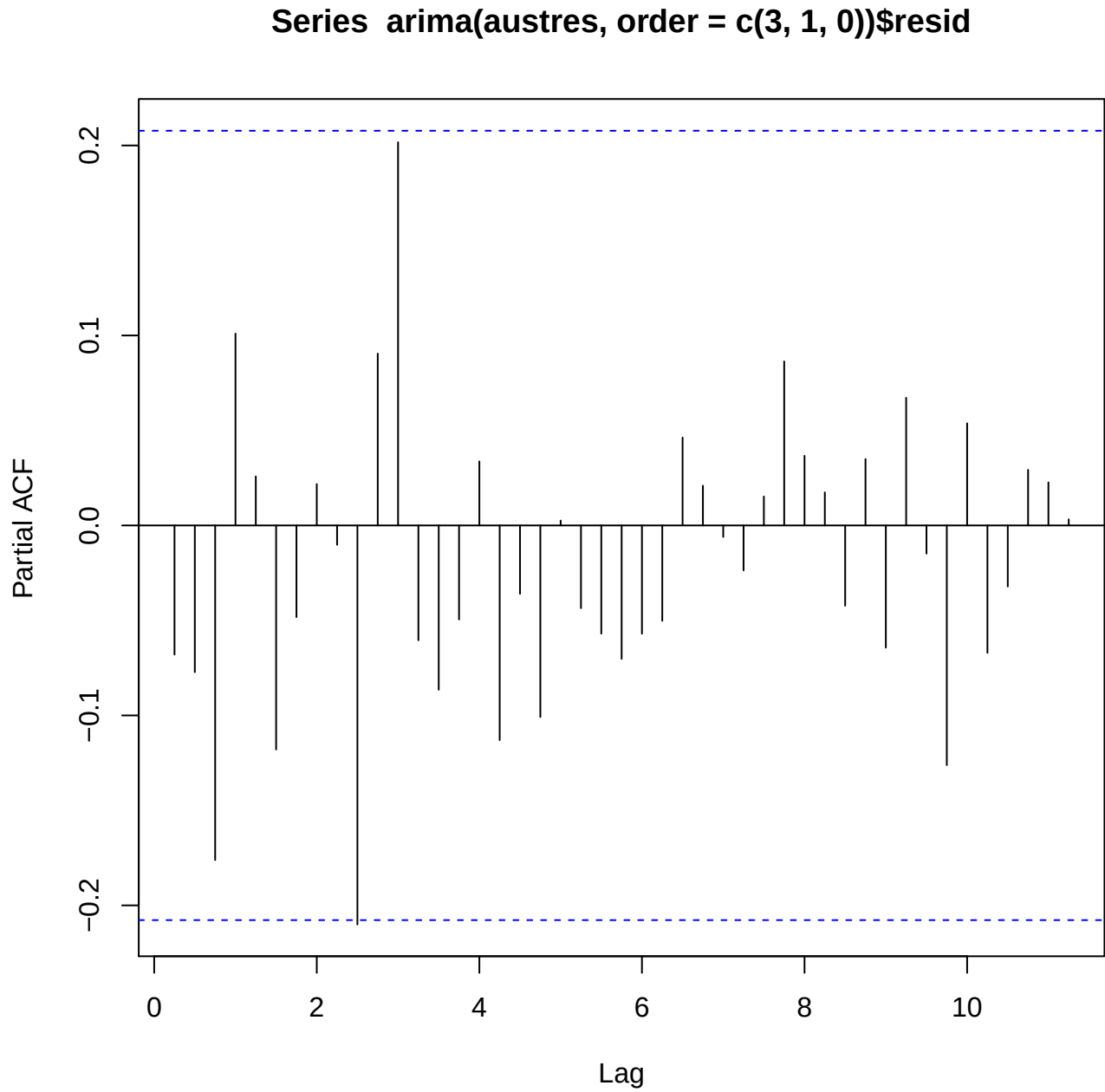
```
> acf(arima(austres, order=c(3,1,0))$resid, lag.max = 45)
> pacf(arima(austres, order=c(3,1,0))$resid, lag.max = 45)
```

The partial correlogram of the residuals for the initial fit shows a very slight significant lag at lag 10. If we are happy to accept this as white noise, then we can stop here.

On the hand, we can try to add 1 moving average term and see what happens.

```
> acf(arima(austres, order=c(3,1,1))$resid, lag.max = 45)
> pacf(arima(austres, order=c(3,1,1))$resid, lag.max = 45)
```

Figure 2: Partial Correlogram of ARIMA(3,1,0) Residuals



The partial correlogram of the ARIMA(3,1,1) model (see Figure 3) is certainly alot better and both ACF and PACF resemble that of white noise.

Now we get the estimates of the parameters of the ARIMA(3,1,1) model for x_t .

```
> arima(austres, order=c(3,1,1))
```

Call:

```
arima(x = austres, order = c(3, 1, 1))
```

Coefficients:

	ar1	ar2	ar3	ma1
	0.9160	-0.1639	0.2437	-0.4358
s.e.	0.2012	0.1699	0.1354	0.1919

sigma^2 estimated as 97.39: log likelihood = -328.25, aic = 666.51

Note that the AIC value of this model is also smaller than that of the ARIMA(3,1,0) model.

So this equation of this model is

$$y_t = 0.9160y_{t-1} - 0.1639y_{t-2} + 0.2437y_{t-3} + w_t - 0.4358w_{t-1},$$

which is

$$\nabla x_t = 0.9160\nabla x_{t-1} - 0.1639\nabla x_{t-2} + 0.2437\nabla x_{t-3} + w_t - 0.4358w_{t-1},$$

where w_t are normally distributed with mean 0 and variance 97.39.

Suppose we want to forecast ahead for the next 6 quarters, then

```
> predict(arima(austres, order=c(3,1,1)), n.ahead=6)
```

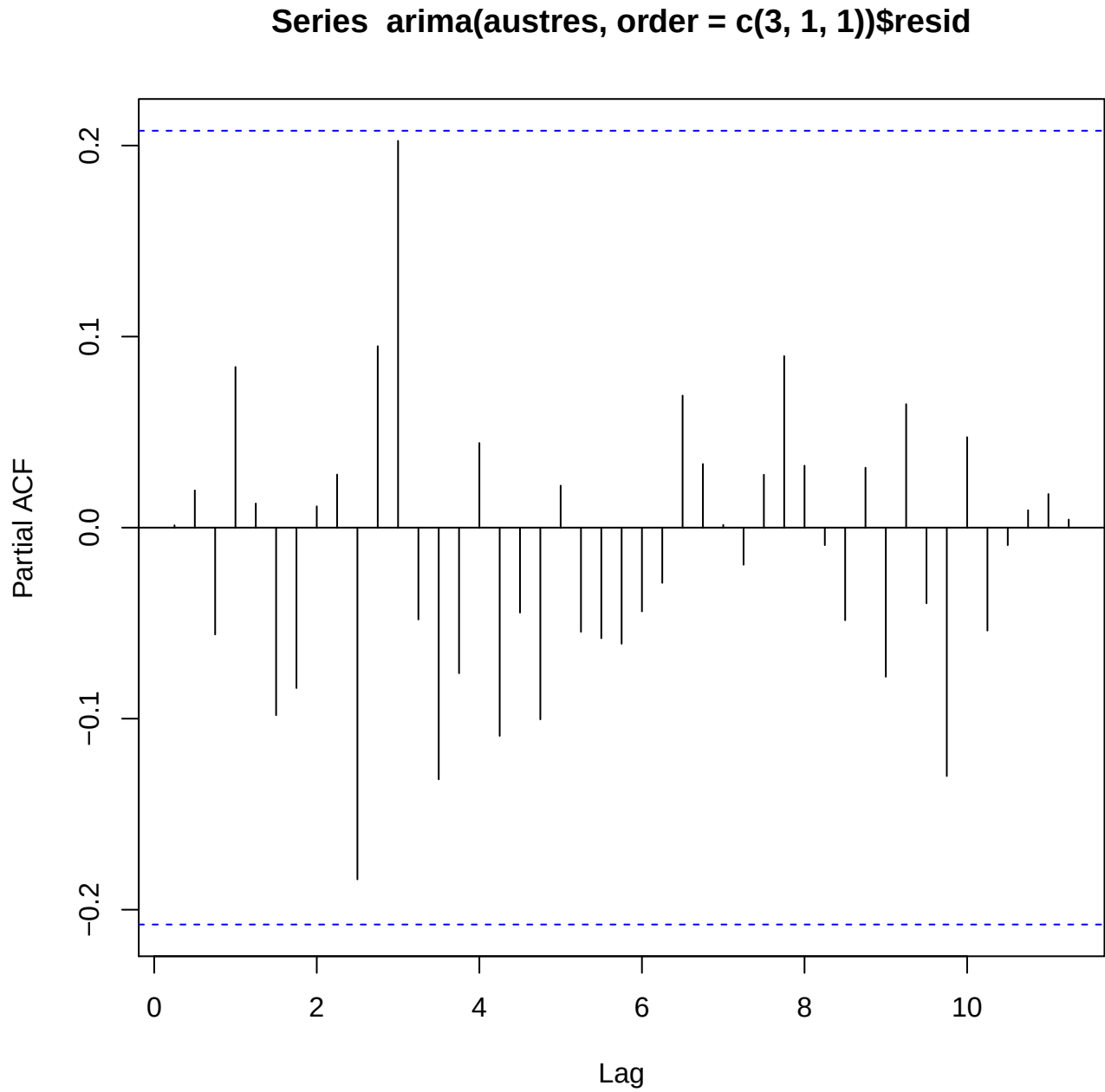
\$pred

	Qtr1	Qtr2	Qtr3	Qtr4
1993			17700.40	17744.62
1994	17787.14	17828.31	17869.83	17911.48

\$se

	Qtr1	Qtr2	Qtr3	Qtr4
1993			9.868598	17.628077
1994	24.719943	32.730365	41.761006	51.438344

Figure 3: Partial Correlogram of ARIMA(3,1,1) Residuals



5. **Example 2.** Airmiles Data. Passenger Revenue Airmiles in U.S. from 1937 to 1960.

We do the plot of the data and see that there is an exponential trend. Next we take the logarithm, but still it looks like there is a increasing trend, so we try taking 1st order difference of the logarithm again to get $\nabla \log x_t$, which is the same as $\log x_t - \log x_{t-1}$.

```
> plot(log(airmiles))
> plot(diff(log(airmiles)))
```

The plot of the first order difference of the logarithm is shown in Figure 3.

Next we look at the correlogram of $\nabla \log x_t$.

```
> acf(diff(log(airmiles)),lag.max=25)
```

The correlogram shows significant correlation of 1 at lag 0, but lags 5 and 6 are close to significant as well, so this is not quite the correlogram of a white noise process. Nevertheless, we try fitting an ARIMA(0,1,0) model to $\log x_t$ first to note its AIC value and to look at its residuals.

```
> arima(log(airmiles),order=c(0,1,0))
```

Call:

```
arima(x = log(airmiles), order = c(0, 1, 0))
```

```
sigma^2 estimated as 0.0566:  log likelihood = 0.39,  aic = 1.22
```

```
> acf(arima(log(airmiles),order=c(0,1,0))$resid)
```

```
> pacf(arima(log(airmiles),order=c(0,1,0))$resid)
```

The correlogram and partial correlogram do not resemble that of a white noise process. The partial correlogram shows a significant lag at lag 2. So we try adding a second order moving average component to the model.

Figure 4: Plot of 1st Order Difference Log Airmiles




```

> arima(log(airmiles),order=c(0,1,2))

Call:
arima(x = log(airmiles), order = c(0, 1, 2))

Coefficients:
          ma1      ma2
      0.7744  0.3679
s.e.  0.2398  0.2584

sigma^2 estimated as 0.02549:  log likelihood = 9.23,  aic = -12.45
> acf(arima(log(airmiles),order=c(0,1,3))$resid, lag.max=30)

```

We see that the AIC value is brought down to -12.45 and the correlogram of the residuals also looks like that of a white noise process. Thus for an ARIMA(0,1,2) model for $\log x_t$, the model is

$$\nabla \log x_t = w_t + 0.7744w_{t-1} + 0.3679w_{t-2},$$

which is the same as

$$\log x_t = \log x_{t-1} + w_t + 0.7744w_{t-1} + 0.3679w_{t-2}.$$

Just for interest if we try fitting an ARIMA(0,1, q) model where q is some other value e.g., $q = 3$ or $q = 4$, it appears that the ARIMA(0,1,3) model has the smallest AIC value. However, given that the correlogram of the residuals of the ARIMA(0,1,2) model already looks like that of a white noise process, the added complexity is probably not needed since there is only a slight decrease in the AIC value.

If we want to forecast ahead the values for the next 5 years say, then

```

> predict(arima(log(airmiles),order=c(0,1,2)), n.ahead=5)
$pred
Time Series:
Start = 1961
End = 1965
Frequency = 1
[1] 10.33755 10.31362 10.31362 10.31362 10.31362

```

```

$se
Time Series:
Start = 1961
End = 1965
Frequency = 1
[1] 0.1596418 0.3251626 0.4719117 0.5828126 0.6757517

> exp(predict(arima(log(airmiles),order=c(0,1,2)), n.ahead=5)$pred)
Time Series:
Start = 1961
End = 1965
Frequency = 1
[1] 30870.42 30140.49 30140.49 30140.49 30140.49

```

6. **Example 3.** The `freeny.y` data. The data is recorded in quarters from 2nd quarter 1962 to 4th quarter 1971. A plot reveals that it looks linearly increasing. So we take first order difference $\nabla y_t = y_t - y_{t-1}$ and also check the correlogram and partial correlogram.

```

> plot(freeny.y)
> plot(diff(freeny.y))
> plot(freeny.y)
> plot(diff(freeny.y))

```

The partial correlogram of the 1st order difference ∇y_t show that it is significant to lag 3. Thus we can try fitting an AR(3) model to ∇y_t , or in other words an ARIMA(3,1,0) model to y_t .

```

> arima(freeny.y, order=c(3,1,0))

```

```

Call:
arima(x = freeny.y, order = c(3, 1, 0))

```

```

Coefficients:
          ar1          ar2          ar3
      0.1067   0.5308   0.1628
s.e.  0.1561   0.1307   0.1575

```

```

sigma^2 estimated as 0.0005437:  log likelihood = 88.37,  aic = -168.74

```

We check the correlogram and partial correlogram of the residuals of this model.

```
> acf(arima(freeny.y, order=c(3,1,0))$resid, lag.max=30)
> pacf(arima(freeny.y, order=c(3,1,0))$resid, lag.max=30)
```

The partial correlogram of the residuals still do not look like white noise PACF and suggests that the residuals may be significantly partially correlated up to lag 2. So we try ARIMA(3,1,2) first and see if we can reduce further. We seem to get an error message with ARIMA(3,1,2), so we try ARIMA(3,1,1) which gives us residuals that resemble white noise.

```
> arima(freeny.y, order=c(3,1,1))
```

Call:

```
arima(x = freeny.y, order = c(3, 1, 1))
```

Coefficients:

	ar1	ar2	ar3	ma1
	0.6227	0.3803	-0.0031	-0.9876
s.e.	0.1707	0.1799	0.1687	0.0419

sigma^2 estimated as 0.0003648: log likelihood = 94.17, aic = -178.33

```
> pacf(arima(freeny.y, order=c(3,1,1))$resid, lag.max=30)
> acf(arima(freeny.y, order=c(3,1,1))$resid, lag.max=30)
```

The ARIMA(3,1,1) model also gives us smaller AIC value (that is, more negative) than the one for ARIMA(3,1,0).

Thus the equation of the model is

$$(1 - 0.6227B - 0.3803B^2 + 0.0031B^3)\nabla y_t = (1 - 0.9876B)w_t$$

which yields

$$\nabla y_t = 0.6227\nabla y_{t-1} + 0.3802\nabla y_{t-2} - 0.0031\nabla y_{t-3} + w_t - 0.9876w_{t-1}$$

$$y_t = y_{t-1} + 0.6227(y_{t-1} - y_{t-2}) + 0.3802(y_{t-2} - y_{t-3}) - 0.0031(y_{t-3} - y_{t-4}) + w_t - 0.9876w_{t-1}.$$

If we want to forecast ahead for the next 8 quarters, then

```
> predict(arima(freeny.y, order=c(3,1,1)), n.ahead=8)
```

```
$pred
```

	Qtr1	Qtr2	Qtr3	Qtr4
1972	9.823412	9.848679	9.875450	9.901641
1973	9.928054	9.954381	9.980741	10.007087

```
$se
```

	Qtr1	Qtr2	Qtr3	Qtr4
1972	0.01924976	0.02295878	0.02777561	0.03153857
1973	0.03523408	0.03865730	0.04195831	0.04513438